

Networks Decide for Us

When homophily governs the criticality of social adoption

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The question

- Collective behaviors (innovations, mobilizations) spread as **adoption cascades** on social networks.
- Most models use *synthetic* topologies and treat agents as cognition-free relays [1].
- Real networks are shaped by **multidimensional homophily** — similar people cluster [2].

Conditional on *fixed* individual propensities, does empirical homophilic structure act as an **autonomous causal force** — shaping not just *how much* adoption occurs, but the **physics of its tipping points**?

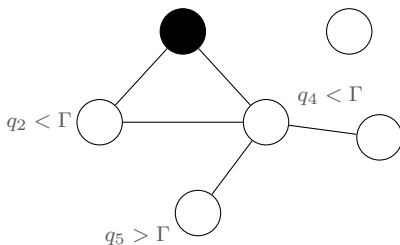
The model

The model — 1) Rational Choice

An innovation has an *Intrinsic Utility Level* (IUL) $\Gamma \in [0, 1]$. Each agent has a *Minimum Utility Requirement* (MUR) $q_i \in [0, 1]$ (its aversion).

An agent adopts on its own iff

$$q_i \leq \Gamma \implies a_i = 1$$



Adoption by intrinsic utility is **deterministic** and **topology-blind**.

The model — 2) Selective Social Influence

If intrinsic adoption fails, agents may adopt by peer contagion. Effective exposure:

$$\tilde{E}_i \equiv \frac{\sum_{j \neq i} \mathbf{X}_{ij} \tilde{a}_j}{\sum_{j \neq i} \mathbf{X}_{ij}}, \quad a_i = \begin{cases} 1 & \text{if } \tau_i \leq \tilde{E}_i \\ 0 & \text{otherwise} \end{cases}$$

Only *influential* infected peers count — those socially *close* in Blau space:

$$\tilde{a}_j = 1 \iff a_j = 1 \wedge U_{ij} \leq \sigma(MSP - d_{ij}), \quad U_{ij} \sim \text{Unif}(0, 1)$$

- d_{ij} : social distance between i and j ; $\sigma(z) = 1/(1 + e^{-z/T})$.
- **MSP** (Maximum Social Proximity): how far across demographic lines influence can reach.

Social adoption is **stochastic** (via U_{ij}) and **topology-sensitive**.

Imputing structure & running the model

Imputing a network onto a survey [2,3]

General Social Survey (GSS 2004)

- Representative; demographics (age, sex, educ, race, religion); *ego* network module.

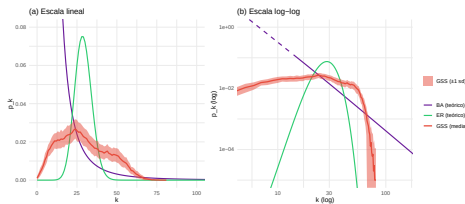
American Trends Panel (ATP 2014)

- Representative; same demographics; *no* network — but rich propensity items (q_i).

Homophily strengths $\hat{\beta}$ from GSS confidant data [4] feed an **ERGM**:

$$\text{logit } \Pr(Y_{ij} = 1) = \alpha + \beta^\top \mathbf{d}_{ij}$$

⇒ plausible discrete topologies ($N = 1000$) on ATP respondents.



Right-skewed degree structure that ER / BA models fail to match jointly.

Comprehensive simulation design

Hybrid social contagion via `netdiffuseR`, over plausible (GSS) vs. **degree-preserving randomized** (GSS-DP) topologies.

- **IUL** (Γ): 41 levels $[0, 1]$. **MSP** (h): 13 levels $[0, 1]$.
- **Thresholds** $\tau_i \sim \mathcal{N}(\mu, \sigma)$: $\mu \in \{.3, .4, .5, .6\}$, $\sigma \in \{.08, .12, .16, .20\}$.
- **Seeding**: random, central, marginal, closeness, eigenvector.

Total: $41 \times 13 \times 4 \times 4 \times 5 \times (\text{runs}) \approx 4 \times 10^6$ **simulations**

We then fit *Big Additive Models* (BAM) to isolate the structural effect:

$$\text{logit}[\mathbb{E}(\Phi)] = \beta_0 + \mathbf{Z}\boldsymbol{\beta} + \beta_{\text{DP}} \mathbb{I}_{\text{DP}} + s(\text{IUL}, \text{MSP})$$

A statistical-physics reading

Mapping social adoption to a ferromagnet

Ising system	NDFU
Spin $s_i \in \{0, 1\}$	$\leftrightarrow$ Adoption state a_i
Magnetization M	$\leftrightarrow$ Adoption $\Phi = N_{\text{ad}}/N$
External field H	$\leftrightarrow$ IUL Γ (rational bias)
Temperature T	$\leftrightarrow$ MSP h (social agitation)
Exchange J_{ij}	$\leftrightarrow$ Homophily / clustering
Susceptibility χ	$\leftrightarrow$ $\text{Var}(\Phi)$ (FDT)

Pseudo-Hamiltonian:

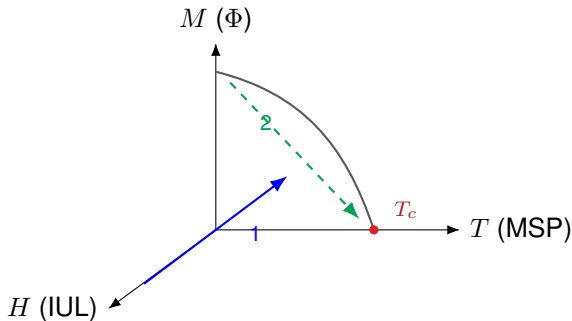
$$\mathcal{H} \approx - \sum_{\langle i,j \rangle} w_{ij}(h) s_i s_j - \Gamma \sum_i s_i$$

Low MSP $\rightarrow$ rigid echo chambers (frozen domains).

High MSP $\rightarrow$ boundaries “melt”, global cascades possible.

Susceptibility diverges near criticality: $\chi = \text{Var}(\Phi) \propto |MSP - MSP_c|^{-\gamma}$

Two paths through the critical point



Path 2 — vary MSP at fixed IUL (“Camino-T”)

- continuous, **2nd-order** transition
- probes the **social channel** $\Rightarrow$ topology-sensitive
- yields susceptibility exponent γ

Path 1 — vary IUL at fixed MSP (“Camino-H”)

- probes the **rational channel** $\Rightarrow$ topology-blind
- yields critical-isotherm exponent δ

Results

Result 1 — A structural premium (volume)

BAM isolates the marginal effect of destroying empirical structure:

$$\beta_{\text{DP}} = -0.850^{***}$$

⇒ randomization cuts adoption odds by ~**57%**

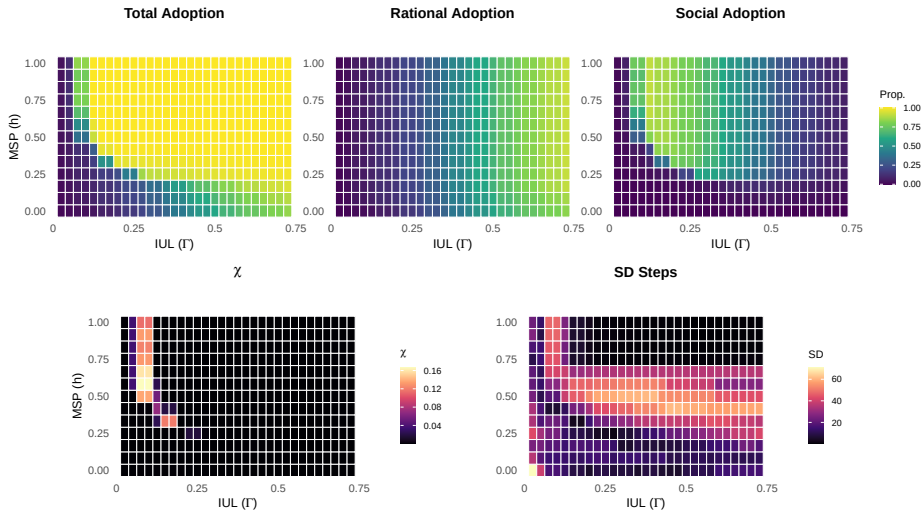
Under *identical* individual propensities, homophilic clustering lets social reinforcement build inside **local echo chambers** and overcome resistance — even for unattractive innovations ($\Gamma < 0.25$).

	Total adoption
τ_{mean}	-6.944***
τ_{sd}	5.942***
DP topology	-0.850***
Dev. explained	82.2%

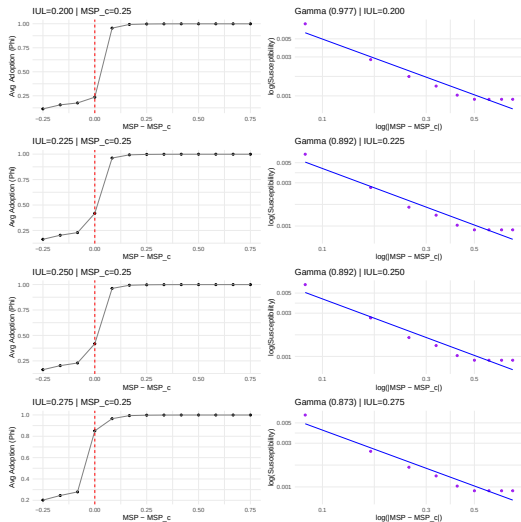
$N = 4,093,440$ simulations.

Result 2 — Criticality signatures (dynamics)

Emergence of Adoption and Phase Transition Signatures



Result 3 — Universality: the discriminator



IUL	GSS γ	GSS-DP γ
0.200	0.98	3.71
0.225	0.89	3.67
0.250	0.89	3.67
0.275	1.54	4.67
mean	0.91	3.93

- GSS: $\gamma \approx 1$ — **Mean-Field universality** (continuous).
- GSS-DP: $\gamma \approx 4$ — no power law; **discontinuous** regime shift.
- Separation $\approx 4.3\times$.

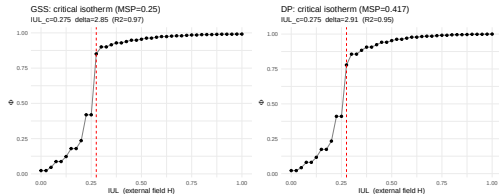
Log-log: $\chi \sim |MSP - MSP_c|^{-\gamma}$ along the Widom line.

Why only the MSP axis discriminates

Two control parameters, two channels:

- **Path 2 (MSP, γ)**: drives the *social* channel.
 $\gamma_{\text{GSS}} \approx 1$ vs. $\gamma_{\text{DP}} \approx 4$ — **topology-sensitive**.
- **Path 1 (IUL, δ)**: drives the *rational* channel
($q_i \leq \Gamma$ is node-local). $\delta_{\text{GSS}} \approx 2.85$, $\delta_{\text{DP}} \approx 2.91$
— **topology-blind** (both $\approx$ MF $\delta = 3$).

The field response reflects the *MUR distribution*, shared by both topologies. The universality **discriminator** must therefore be the MSP (temperature) axis.



Critical isotherms $\Phi(IUL)$: same $\delta \approx 3$ for both.
(exploratory)

The mean-field paradox — and its resolution

Puzzle

Mean-field $\gamma = 1$ usually needs a *well-mixed* system. We recover it in a *strongly clustered, homophilic* network.

Resolution

Homophilic clusters behave as quasi-isolated subsystems coupled by *weak, similarity-selective bridges*. As $MSP \rightarrow MSP_c$ these bridges activate together, synchronizing clusters into one coherent order parameter — *indistinguishable* from mean-field behavior. Randomization (DP) destroys this and yields discontinuity instead.

Homophily provides a “**sociological viscosity**” that converts chaotic jumps into legible cascades.

Conclusions

Conclusions

Networks decide for us — twice over.

1. **Volume:** empirical homophily yields a **structural premium** (+57% odds).
2. **Dynamics:** homophily makes tipping points **continuous & predictable** (mean-field, $\gamma \approx 1$); randomization makes them **abrupt & volatile** ($\gamma \approx 4$).
3. **Mechanism:** the social (MSP) axis carries universality; the rational (IUL) axis is topology-blind ($\delta \approx 3$).

Macroscopic social change is not the linear sum of individual preferences — it is governed by the structure of our ties.

Thanks!

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`github.com/aoliveram/Networks-Decide-for-Us`